Trade Publishing / Reporting Classes

Module Goal: Record trades and make them available to consumers (websocket, logs, etc)

NBBO: Nation Best bid and Offer
Bid -> highest price someone is willing to buy
Offer (Ask) -> lowest price someone is willing to sell

Captures full match details

Struct 3

Trade

+ double Price
+ double Qty
+ Std::Chrono::System_Clock::time_point timeStamp
+ uint64_t makerOrderId
+ uint64_t takerOrderId
+ Std::string Symbol
+ Side takerSide

TradePublisher

- Std::vector<Std::function<void(const Trade&)>> Subscribers
- Std::mutex SubMutex

+ Void PublishTrade (const Trade & trade)
+ Void Subscribe (std::function<void(const Trade&)> Callback

Observer Pattern
MatchEngine publishes to it

- Supports multiple consumers (WebSocket, file, etc)
- Thread-Safe Publish / Subscribe

OrderBook

MatchEngine

Depends on OrderBook for real NBBO per Symbol

The System calculates the last traded price and the best buy and sell prices for each symbol / Stock

MarketDataAggregator

- Std::unordered_map<Std::string, double> LastTradePrices
- Std::unordered_map<Std::string, Std::pair<double, double>> nbbo
- Std::mutex aggMutex

+ Void updateFromTrade (const Trade& trade)
+ double getLastTradPrice (const Std::string& Symbol)
+ Std::pair<double, double> getNBBO (const Std::string & Symbol)